

Module 2

Accounting Statements and Finance

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

Why Do We Need Accounting?

1. What do firms own?
 - Real assets (tangible, intangible) and financial assets (cash, securities)

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2. What do firms owe?

Why Do We Need Accounting?

1. What do firms own?
 - Real assets (tangible, intangible) and financial assets (cash, securities)
2. What do firms owe?
3. How much money did firms make?
 - Income? Cash flow?

Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?



CNBC, August 18, 2026 (3:50),
“Anthropic tells investors annualized
revenue run rate climbed to \$65
billion in July”

Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?
- We will have a clear understanding after this module



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Three Financial Statements

- **Balance sheet**
 - What do firms own and owe at a point in time



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- **Balance sheet**
 - What do firms own and owe at a point in time
- **Income statement**
 - How much did firms earn in the past year



Three Financial Statements

- **Balance sheet**
 - What do firms own and owe at a point in time
- **Income statement**
 - How much did firms earn in the past year
- **Statement of cash flows**
 - Cash inflows and outflows during the past year



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- **SEC**, the federal agency that regulates U.S. securities markets and enforces disclosure



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- **10-K**, a firm's audited annual report filed with the SEC



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Goals

- Interpret the balance sheet, income statement, and statement of cash flows

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- Distinguish between market and book values

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- Distinguish between market and book values
- Explain why income differs from cash flow
- Understand the taxation of corporate and personal income

Module 2, Part 1

The Balance Sheet

Balance Sheet

- **Balance sheet**, assets, liabilities, and equity at a point in time

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 - **Shareholders' equity**, the residual claim on assets after all liabilities are settled

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$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

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Where does the balance sheet come from?

- Recall from **Module 1**, businesses sell financial assets (financing) to purchase real assets (investment)

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- The numbers are the cost of acquiring real assets, or the amount raised from selling financial assets
- The numbers stay at the transaction amount, even when the asset is worth more or less later

Example, Issuing Stock to Build a Data Center (\$ Billions)

1. Today

Cash	1	Liabilities	0
Fixed assets	0	Shareholders' equity	1
Total assets	1	Total liabilities and equity	1

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Breaking Down the Balance Sheet of a Business (\$ Millions)

Current assets

Assets that can generate return within one year

Assets

Current assets

Cash and marketable securities	2,600
Accounts receivable	500
Inventories	9,000
Total current assets	12,100

Fixed assets

Tangible fixed assets

Property, plant, equipment	48,200
Less accumulated depreciation	19,700
Net tangible fixed assets	28,500

Intangible asset	700
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Total fixed assets	29,200
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Total assets	41,300
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Liabilities and equity

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Accounts payable	9,900
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Long-term debt	11,300
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Paid-in capital	6,300
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Breaking Down the Balance Sheet of a Business (\$ Millions)

Most liquid 	Assets		Liabilities and equity		Paid first 
	Current assets		Current liabilities		
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	Accounts receivable	500	Accounts payable	9,900	
	Inventories	9,000	Total current liabilities	10,100	
	Total current assets	12,100	Long-term debt	11,300	
	Fixed assets		Total liabilities	21,400	
	Tangible fixed assets		Shareholders' equity		
	Property, plant, equipment	48,200	Paid-in capital	6,300	
	Less accum. depreciation	19,700	Retained earnings	13,600	
Net tangible fixed assets	28,500	Total shareholders' equity	19,900		
Intangible asset	700			Paid last	
Total fixed assets	29,200				
Least liquid	Total assets	41,300	Total liabilities and equity	41,300	

- **Liquidity**, the ease of converting an asset into cash

Book Value vs. Market Value

1. **Book values**, the cost of acquiring real assets, or the amount raised from selling financial assets
 - Values on balance sheets are book value

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2. **Market values**, what things would sell for today
 - E.g., worth \$4 billion today → market value

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 - Values on balance sheets are book value
 - E.g., the \$2 billion data center built last year → book value, forever
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 - E.g., worth \$4 billion today → market value
- In general, book $\neq$ market value

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- *Should you revise shareholders' equity if someone is willing to buy the business at \$4 billion? **No***

Book Value vs. Market Value

1. Book value of equity = Book value of assets – Book value of liabilities
 - E.g., Apple at the end of fiscal year 2025,
 - Book value of assets, \$359,241 Mn
 - Book value of liabilities, \$285,508 Mn
 - What is the book value of equity?

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 - What is the book value of equity?
 - Answer, $\$359,241 \text{ Mn} - \$285,508 \text{ Mn} = \$73,733 \text{ Mn}$

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 - Answer, $\$359,241 \text{ Mn} - \$285,508 \text{ Mn} = \$73,733 \text{ Mn}$
2. Market value of equity = Market price of each share $\times$ Total shares outstanding
 - E.g., Apple at the end of 2025,
 - Share price, about \$271
 - Shares outstanding, 14,800 Mn
 - What is the market value of equity?

Book Value vs. Market Value

1. Book value of equity = Book value of assets – Book value of liabilities
 - E.g., Apple at the end of fiscal year 2025,
 - Book value of assets, \$359,241 Mn
 - Book value of liabilities, \$285,508 Mn
 - What is the book value of equity?
 - Answer, $\$359,241 \text{ Mn} - \$285,508 \text{ Mn} = \$73,733 \text{ Mn}$
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 - What is the market value of equity?
 - Answer, $\$271 \times 14,800 \text{ Mn} = \$4,010,800 \text{ Mn}$

Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm



Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm
- Market value of equity reflects the money investors expect the firm to make in the future



Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm
- Market value of equity reflects the money investors expect the firm to make in the future
- Naturally, market value of equity $>$ book value of equity for a successful business



Practice Problem

- A business's book value of total assets is \$45,000. Its book value of current liabilities is \$15,000, and its book value of shareholders' equity is \$20,000. How much is its book value of long-term liabilities?

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- $\$45,000 = \$15,000 + \$10,000 + \$20,000$

Quick Summary

- Accounting conveys information to investors and stakeholders
- The balance sheet, what a company owns and how it financed it
- Left side (assets) must equal right side (liabilities + equity)
- Market and book values generally differ

Module 2, Part 2

The Income Statement

Income Statement

- **Income statement**, how much firms earned in the past year



INCOME STATEMENTS

Income Statement

- **Revenues**, sales made in the period, whether or not cash was collected
 - E.g., a firm sells 100 products to customers, each at a price of \$100
 - $\text{Revenues} = 100 \times \$100 = \$10,000$

	\$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	5,000
Interest expense	1,000
Taxable income	4,000
Taxes	840
Net income	3,160
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Income Statement

- **Operating expenses**, the costs of running the business

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Income Statement

- **Operating expenses**, the costs of running the business
 - **Cost of goods sold (COGS)**, the direct cost of the goods actually sold
 - E.g., each of the 100 products sold costs the firm \$60 to make
 - $COGS = 100 \times \$60 = \$6,000$

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 - **Selling, general & administrative (SG&A)**, other operating costs, e.g., marketing, salaries, distribution
 - **Depreciation**, this period's share of a fixed asset's cost

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- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once



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- Two things happen with depreciation,
 - In the income statement, subtract depreciation from revenues as a non-cash expense



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- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once
- This spreading is **depreciation**
- Two things happen with depreciation,
 - In the income statement, subtract depreciation from revenues as a non-cash expense
 - In the **balance sheet**, add to accumulated depreciation



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Cash flow (\$ billions)	-2.0	0	0	0	0	0	
Effect on net income	0	-0.4	-0.4	-0.4	-0.4	-0.4	← Depreciation

Income Statement

- **EBIT**, earnings before interest and income taxes
 - $\text{EBIT} = \text{Revenues} - \text{COGS} - \text{SG\&A expenses} - \text{Depreciation}$

	\$ Million
Revenues	80,000
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Allocation of net income	
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Income Statement

- **Interest expense**, the interest paid on the firm's debt for the period

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Income Statement

- **Taxes**

- Taxable income = EBIT – Interest expense
- Taxes = Taxable income × Tax rate

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Where Net Income Goes

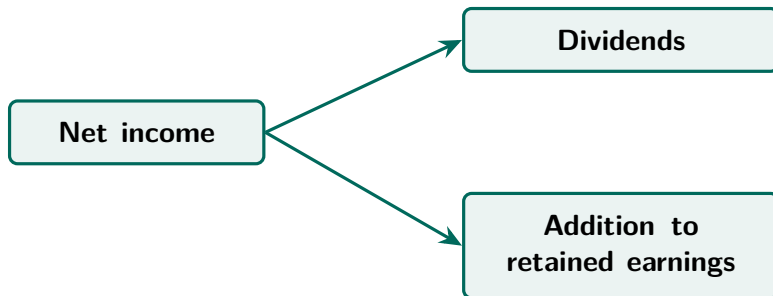
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- Net income can be kept in the business, or paid out to shareholders

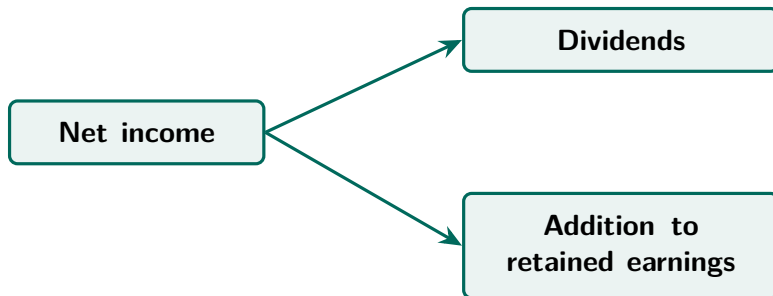
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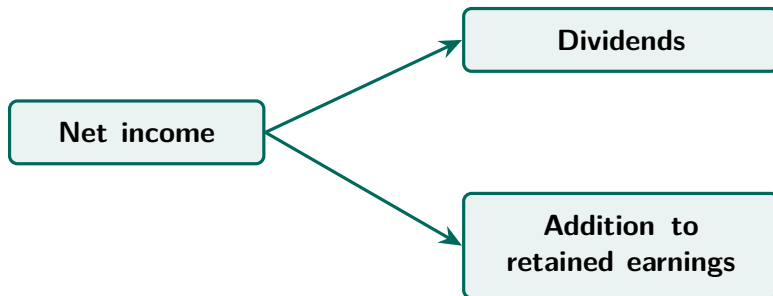
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- **Dividends**, cash paid out to shareholders now
- **Addition to retained earnings**, profit kept in the business and reinvested on the shareholders' behalf

Income Statement, A Memory Aid

Line item	Letters	Remember it as
Revenues	R	R ainbow
COGS	C	C andy
SG&A expenses	S	S ells
Depreciation	D	D ependably
EBIT	E	E specially
Interest expense	I	I n
Taxable income	T.I.	T exas I ncorporated
Taxes	T	T exas
Net income	N.I.	N ever I ncorporates
Dividends	D	D uring
Addition to retained earnings	R.E.	R e- E lections

Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?



CNBC, August 18, 2026 (3:50),
“Anthropic tells investors annualized
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Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?
- Profitable on an **EBITDA** basis
 - EBITDA, earnings before interest, taxes, depreciation, and amortization
 - Amortization, the same cost-spreading as depreciation, applied to intangible assets



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- Profitable on an **EBITDA** basis
 - EBITDA, earnings before interest, taxes, depreciation, and amortization
 - Amortization, the same cost-spreading as depreciation, applied to intangible assets
- Not profitable on a **net income** basis



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Practice Problem

	\$
Revenues	?
COGS	20,000
SG&A expenses	10,000
Depreciation	8,000
EBIT	12,000
Interest expense	?
Taxable income	?
Taxes (tax rate = 21%)	2,100
Net income	7,900
Dividends	2,500
Addition to retained earnings	?

Practice Problem

	\$
Revenues	50,000
COGS	20,000
SG&A expenses	10,000
Depreciation	8,000
EBIT	12,000
Interest expense	2,000
Taxable income	10,000
Taxes (tax rate = 21%)	2,100
Net income	7,900
Dividends	2,500
Addition to retained earnings	5,400

Quick Summary

- The income statement, what a company accomplished over a period
- The order of line items matters greatly
- Mixing it up has drastic consequences
 - Taxes should always be computed **after** depreciation and interest, never before

Module 2, Part 3

The Statement of Cash Flows

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- **Operating activities**, cash from running the business day to day
- **Investing activities**, cash spent buying fixed assets and cash received from selling them
- **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

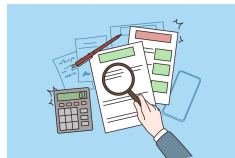
Statement of Cash Flows

Direct method	Indirect method
Lists every cash receipt and payment	Starts from net income and adjusts it
More time-consuming to prepare	Faster to prepare
Rarely used in practice	The common choice

Direct →



Indirect →



Statement of Cash Flows

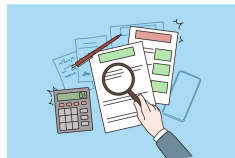
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- Both methods give the same totals

Direct →



Indirect →



Statement of Cash Flows

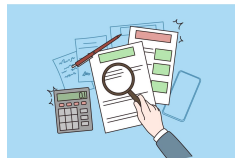
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More time-consuming to prepare	Faster to prepare
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- Both methods give the same totals
- We will follow the indirect method, starting from net income in the income statement

Direct →



Indirect →



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Effect on net income	0	-0.4	-0.4	-0.4	-0.4	-0.4

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Relationship of cash flow (CF) and net income (NI)?

When spending on fixed assets, $CF = NI - \text{spending on fixed assets}$

When recording depreciation, $CF = NI + \text{depreciation}$

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- The firm issues (sells) \$1 billion of new stock to investors

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When raising money, $CF = NI + \text{amount raised}$

How Do Net Income and Cash Flow Differ?

- The firm spends \$60 million producing AI control panels
- None of them have been sold yet



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

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Cash flow (\$ millions)	−60
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<u>Change in inventories</u>	<u>+60</u>



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When adding to inventories, $CF = NI - \text{change in inventories}$



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- Net income rises now, cash flow comes later when customers actually pay



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When firm pays off suppliers, $CF = NI + \text{change in accounts payable}$



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Operating Activities

Operating activities, cash from running the business day to day

Steps to calculate cash flow from operating activities,

1. **Start** with net income
2. **Add** depreciation
3. **Subtract** change in net working capital (NWC)
 - $\text{NWC} = \text{inventories} + \text{accounts receivable} - \text{accounts payable}$
 - NWC, cash tied up in day-to-day operations, in unsold goods and unpaid bills

Why Subtract Change in Net Working Capital (NWC)?

When adding to inventories, $CF = NI - \text{change in inventories}$

When firm not yet paid by customers, $CF = NI - \text{change in accounts receivable}$

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Ignoring other adjustments,

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Ignoring other adjustments,

$$CF = NI$$

- change in inventories*
- change in accounts receivable*
- + change in accounts payable*

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Ignoring other adjustments,

$$CF = NI$$

– change in inventories

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$$CF = NI - \text{change in (inventories + accounts receivable – accounts payable)}$$

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Ignoring other adjustments,

$$CF = NI$$

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$$CF = NI - \text{change in (inventories + accounts receivable – accounts payable)}$$

$$CF = NI - \text{change in net working capital (NWC)}$$

Operating Activities

Cash flow from operating activities	
Net income	3,160
Depreciation	3,000
Changes in working capital items	−1,000
Cash flow from operating activities	<u>7,160</u>

Investing Activities

Investing activities, cash spent buying fixed assets and cash received from selling them

Investing activities show cash flows from,

1. Capital purchased and sold

- **Capital**, a firm's fixed assets
- e.g., machinery, plant, land
- **Capital expenditure**, the amount spent on buying capital during the period
- Cash flow is *negative* if firms buy capital

Investing Activities

Cash flow from investing activities	
Capital expenditure	−3,000
Sales of fixed assets	200
Cash flow from investing activities	−2,800

Financing Activities

Financing activities, cash raised from lenders and shareholders, and cash returned to them

Financing activities show cash flows from,

1. Selling new shares → Cash ↑
2. Paying dividends → Cash ↓
3. Taking new debt, either a loan or a bond → Cash ↑
4. Repaying principal amount of old debt → Cash ↓

Quick Summary

- The statement of cash flows, the company's cash inflows and outflows
- Cash flow from operating activities
 - $\text{Net income} + \text{depreciation} - \text{change in NWC}$
- Cash flow from investing activities
- Cash flow from financing activities

Practice Problem

- Which of the following will make cash flow higher than net income for a firm?
 - a. Increase in inventories.
 - b. Decrease in accounts payable.
 - c. Decrease in accounts receivable.
 - d. Expenditures on fixed assets.

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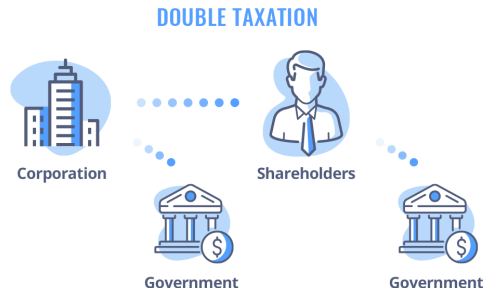
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- Cash flow from operating activities = Net income + Depreciation – Change in NWC
- $\$3,160 + \$3,000 - (-\$1,000) = \$7,160$ million

Module 2, Part 4

Taxation

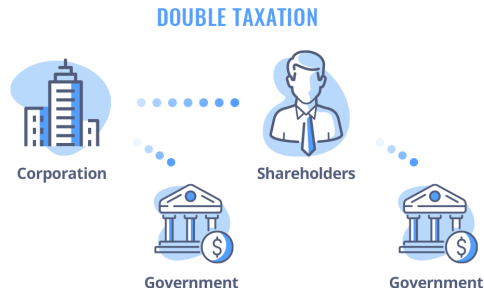
Corporate Taxation, Double Taxation

- Recall from **Module 1**,
 - The corporation pays tax on its income



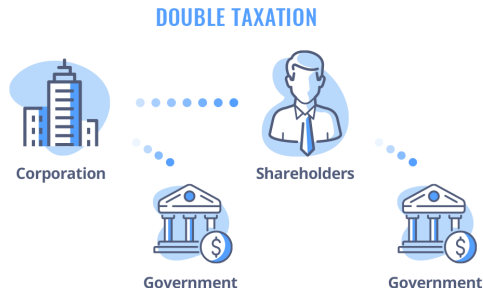
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- Recall from **Module 1**,
 - The corporation pays tax on its income
 - Shareholders pay tax again on what they receive



Corporate Taxation, Double Taxation

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 - The corporation pays tax on its income
 - Shareholders pay tax again on what they receive
- This is **double taxation**



Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is **21%**



WSJ, February 28, 2025 (6:57),
“Economists on How Trump’s 2017
Tax Cuts Actually Played Out”

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- The rate is a major political topic
 - Higher rate, smaller deficit, more funding for public services



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- **One Big Beautiful Bill Act (2025)**, extended and made permanent much of it



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Corporate Taxation, Tax Shields

- Depreciation and interest come first

Line item	Letters	Remember it as
Revenues	R	R ainbow
COGS	C	C andy
SG&A expenses	S	S ells
Depreciation	D	D ependably
EBIT	E	E specially
Interest expense	I	I n
Taxable income	T.I.	T exas I ncorporated
Taxes	T	T exas
Net income	N.I.	N ever I ncorporates
Dividends	D	D uring
Addition to retained earnings	R.E.	R e- E lections

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- Taxes come last
- Depreciation and interest reduce corporate taxes

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Corporate Taxation, Tax Shields

- Interest expense can
 - act as a “shield” against corporate taxation
 - increase the *total* cash flow to investors (debt and equity investors combined)

A. Has debt

EBIT	\$5,000	
Interest expense	\$1,000	→ debt investors
Taxable income	\$4,000	
Taxes (tax rate = 21%)	\$840	
Net income	\$3,160	→ equity investors

- Investors get \$4,160 in total with debt, \$3,950 without

B. No debt

EBIT	\$5,000	
Interest expense	\$0	→ debt investors
Taxable income	\$5,000	
Taxes (tax rate = 21%)	\$1,050	
Net income	\$3,950	→ equity investors

Corporate Taxation, Tax Shields

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- **Interest tax shield**, the tax saving created by deducting interest before taxes
- Interest tax shield = Tax rate $\times$ Interest expense
 - Each dollar of interest lowers taxable income by \$1, so taxes fall by the tax rate $\times$ \$1
 - In the example, $21\% \times \$1,000 = \210 , exactly the extra amount investors get with debt (\$4,160 – \$3,950)

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 - Each dollar of interest lowers taxable income by \$1, so taxes fall by the tax rate $\times$ \$1
 - In the example, $21\% \times \$1,000 = \210 , exactly the extra amount investors get with debt (\$4,160 – \$3,950)
- Should a firm take on as much debt as possible? No (bankruptcy costs).

Corporate Taxation, Tax Shields

- Depreciation can also
 - act as a “shield” against corporate taxation
 - increase the total cash flow to investors

Income Statement (\$ Million)

Baseline	
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	5,000
Interest expense	1,000
Taxable income	4,000
Taxes	840
Net income	3,160

Corporate Taxation, Tax Shields

- Depreciation can also
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 - increase the total cash flow to investors
- Example, Suppose a business has an *additional* \$1,000 million in depreciation, and its tax rate is 21%

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 - increase the total cash flow to investors
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Income Statement (\$ Million)		
	Baseline	More depreciation
Revenues	80,000	80,000
COGS	56,000	56,000
SG&A expenses	16,000	16,000
Depreciation	3,000	4,000
EBIT	5,000	4,000
Interest expense	1,000	1,000
Taxable income	4,000	3,000
Taxes	840	630
Net income	3,160	2,370

Corporate Taxation, Tax Shields

- Questions,
 - What would be the impact on net income?
 - What would be the impact on cash flow?

\$ Million	Baseline	More depreciation
Net income	3,160	2,370
Add back depreciation	3,000	4,000
Cash flow	6,160	6,370

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- Questions,
 - What would be the impact on net income?
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Add back depreciation	3,000	4,000
Cash flow	6,160	6,370

- Answers,
 - Net income *falls* by \$790 million
 - Cash flow *rises* by $21\% \times \$1,000 \text{ million} = \210 million

Corporate Taxation, Tax Shields

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Add back depreciation	3,000	4,000
Cash flow	6,160	6,370

- Answers,
 - Net income *falls* by \$790 million
 - Cash flow *rises* by $21\% \times \$1,000 \text{ million} = \210 million
- Depreciation cuts taxes, reducing net income but raising cash flow
 - Cash flow increases by $\text{Tax rate} \times \$1$ if depreciation is \$1 higher

Practice Problem

- A business pays a 21% tax rate. Suppose it had an additional \$200 million in depreciation. By how much would its taxes fall? What about its total cash flow to investors?

Practice Problem

- A business pays a 21% tax rate. Suppose it had an additional \$200 million in depreciation. By how much would its taxes fall? What about its total cash flow to investors?
- Taxes fall by $21\% \times \$200 \text{ million} = \42 million

Practice Problem

- A business pays a 21% tax rate. Suppose it had an additional \$200 million in depreciation. By how much would its taxes fall? What about its total cash flow to investors?
- Taxes fall by $21\% \times \$200 \text{ million} = \42 million
- Cash flow to investors rises by the same \$42 million, the depreciation tax shield

Personal Taxation

- The following personal tax rates matter for salaried employees and non-corporate businesses

Taxable Income (Single Taxpayers)	Marginal Tax Rate (%)
\$0–\$9,525	10
\$9,526–\$38,700	12
\$38,701–\$82,500	22
\$82,501–\$157,500	24
\$157,501–\$200,000	32
\$200,001–\$500,000	35
\$500,001 and above	37

Personal Taxation, Average vs. Marginal Tax Rates

- Marginal tax rate = Tax rate that is applied to each extra dollar of income
- Average tax rate = Total taxes / Total income

Example, Average Tax Rate

Suppose your salary is \$150,000 per year. What would be your average tax rate according to the given tax schedule for single taxpayers?

Taxable Income (Single Taxpayers)	Marginal Tax Rate (%)
\$0–\$9,525	10
\$9,526–\$38,700	12
\$38,701–\$82,500	22
\$82,501–\$157,500	24
\$157,501–\$200,000	32
\$200,001–\$500,000	35
\$500,001 and above	37

- Total tax = $\$9,525 \times 10\% + (\$38,700 - \$9,525) \times 12\% + (\$82,500 - \$38,700) \times 22\% + (\$150,000 - \$82,500) \times 24\%$
- Average tax rate = $30,289.5 / 150,000 = 20.19\%$

Practice Problem

If a firm's net income is positive and its noncash expenses are positive, which of the following could account for a negative amount of cash provided by operations?

- a. Current assets decrease more than current liabilities decrease.
- b. Current assets increase more than current liabilities increase.
- c. Current assets decrease more than current liabilities increase.
- d. A large addition is made to plant and equipment.

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Practice Problem

Here are the 2021 and 2022 (incomplete) balance sheets for Newble Oil Corporation.

BALANCE SHEET AT END OF YEAR (Figures in \$ millions)					
Assets	2021	2022	Liabilities and Shareholders' Equity	2021	2022
Current assets	\$328	\$510	Current liabilities	\$300	\$258
Net fixed assets	1,380	1,510	Long-term debt	920	1,100

- What was shareholders' equity at the end of 2021 and 2022?
2021, 488 2022, 662
- If Newble paid dividends of \$185 million in 2022 and made no stock issues, what must have been net income during the year? **$174 + 185 = 359$**
- If Newble purchased \$385 million in fixed assets during 2022, what must have been the depreciation charge on the income statement? **$385 - 130 = 255$**

Quick Summary

- Interest and depreciation come before taxes, so both shield income from tax
- A tax shield is worth the tax rate times the deduction, leaving more cash for investors
- Marginal rate applies to each extra dollar of income, average rate is total taxes over total income

Income Statement

	\$ Million	% of Sales
Revenues	80,000	100.0%
COGS	56,000	70.0%
SG&A expenses	16,000	20.0%
Depreciation	3,000	3.8%
EBIT	5,000	6.3%
Interest expense	1,000	1.3%
Taxable income	4,000	5.0%
Taxes	840	1.1%
Net income	3,160	4.0%
Allocation of net income		
Dividends	1,000	1.3%
Addition to retained earnings	2,160	2.7%

Key Takeaways

- Three statements, what a firm owns and owes, what it earned, and how cash moved

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- Book values are the cost of acquiring real assets or the amount raised from selling financial assets, market values are what things would sell for today
- Net income $\neq$ cash flow, add back depreciation, subtract the change in NWC
- Interest and depreciation shield taxes, leaving more cash for investors

Thank you!

Please let me know if you have any questions.